CLIMATE IMPACT AND CORRELATION ANALYZER

Exploring Interrelationships of Climate Indicators through Data Visualization

GROUP 5

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INTRODUCTION

The global climate crisis, characterized by rising temperatures, melting ice caps, and shifting precipitation patterns, has disrupted ecosystems, agriculture, and human health. Increased CO_2 emissions have worsened climate variability, while sea-level rise and more frequent natural disasters highlight the need to understand these interconnected factors. Traditional tools often fail to reveal these complex relationships, underscoring the need for advanced analytical solutions.

To address this, we developed an interactive climate data visualization platform that integrates datasets on surface temperatures, carbon emissions, sea-level rise, and natural disasters. Using Python-based tools like Pandas, NumPy, Plotly, and Chart.js, the platform offers user-friendly interfaces and allows data filtering by country, region, and time. Visualizations such as heatmaps, choropleth maps, and 3D globes provide insights into climate dynamics.

This approach enables users to explore correlations between emissions, temperature anomalies, sea-level trends, and natural disasters. By highlighting these connections, the platform supports evidence-based decision-making, helping researchers, policymakers, and stakeholders develop more effective climate strategies.

RELATED WORK

Previous climate data visualizations mainly include static maps, charts, and single-variable interactive dashboards, such as IPCC summaries [1] and NASA's temperature anomaly maps. Liu et al. [2] and Smith et al. [3] created tools focusing on emissions-to-temperature and emissions-sea-level connections. However, few integrate natural disaster data, potentially overlooking key links between warming and disaster frequency. Our platform builds on these efforts by unifying multiple climate indicators and incorporating disaster data, offering a comprehensive view of climate dynamics and their societal and environmental impacts.

DATASETS AND PREPROCESSING

Our platform integrates temporal and spatial climate datasets from Kaggle and other public archives. These datasets include:

- **Global Surface Temperature:** Monthly average surface temperatures from the 18th century to present, enabling long-term trend analysis of warming patterns.
- **Carbon Emissions:** Country-level carbon dioxide equivalent emissions spanning from the early 20th century through recent decades, allowing the examination of historical emission trajectories.
- **Sea Level Rise Data:** Historical mean sea level (MSL) records extending back to the 19th century, capturing long-term changes associated with glacial melt and thermal expansion.
- **Natural Disaster:** A compilation of events (e.g., floods, droughts, earthquakes, extreme weather) from the early 20th century to the present, permitting correlation of disaster occurrences with temperature anomalies and emissions trends.

Data Cleaning, Transformation, and Exploratory Analysis

We cleaned the data by removing irrelevant columns, handling missing values, and standardizing units (e.g., temperature in Celsius, emissions in metric tons of CO_2-equivalent). Data was aligned to a consistent "Country–Year" format for easy comparison. In the exploratory analysis, we used statistical techniques and preliminary plots to identify key patterns and inform the selection of final visualizations.

IMPLEMENTATION

This is the Homepage of our interface. The page consists of tabs that lets us select what visualization we need.

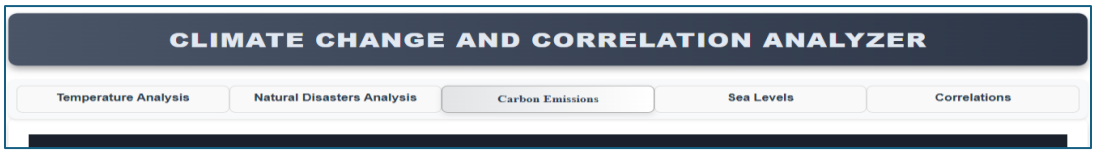


Fig 1: Home Page

Temperature Visualizations:

- *Calendar Heat Maps:* Showing daily temperature variations and seasonal patterns for a selected year and location(*Fig 2*).
- *3D Globe Representation*: A global perspective on temperature disparities, allowing users to rotate, zoom, and inspect different regions(Fig 3).
- *Choropleth Maps:* Displaying spatial patterns of temperature changes over time with adjustable temporal slides(*Fig 4*).

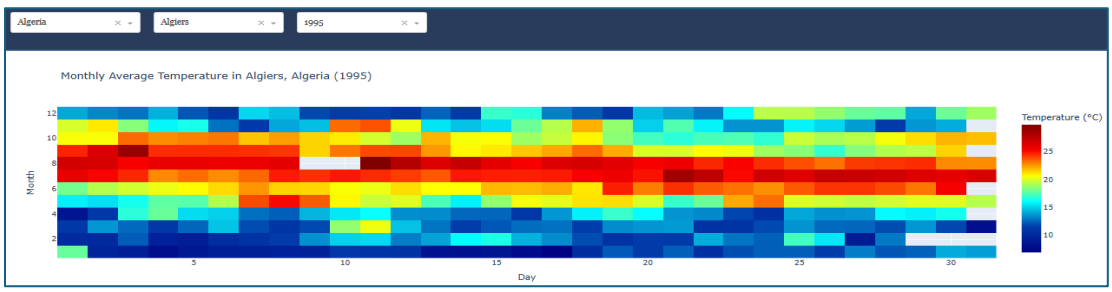


Fig 2: illustrates a calendar heat map for a given city-year combination

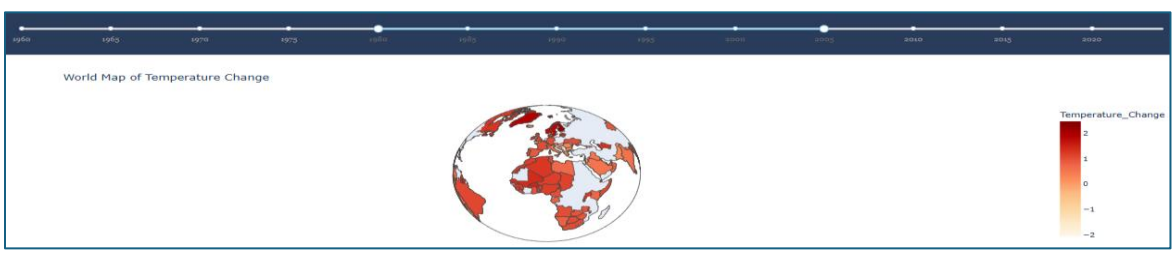


Fig 3: Presents a 3D globe of temperature anomalies with time sliders

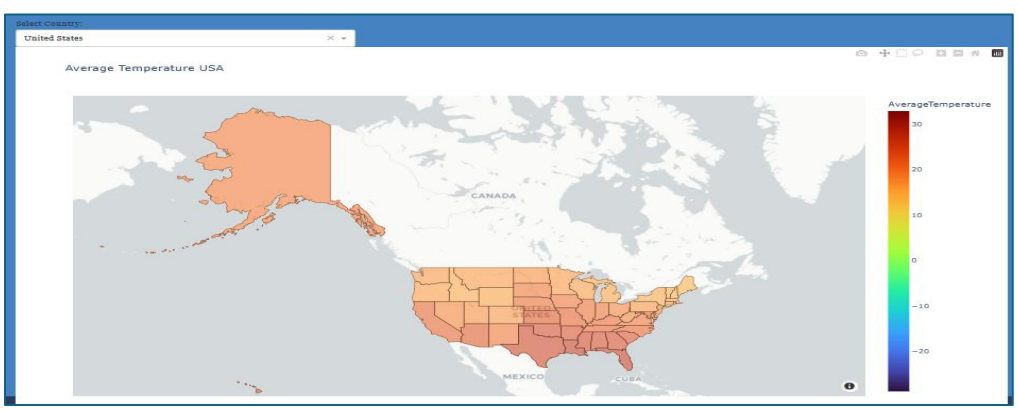
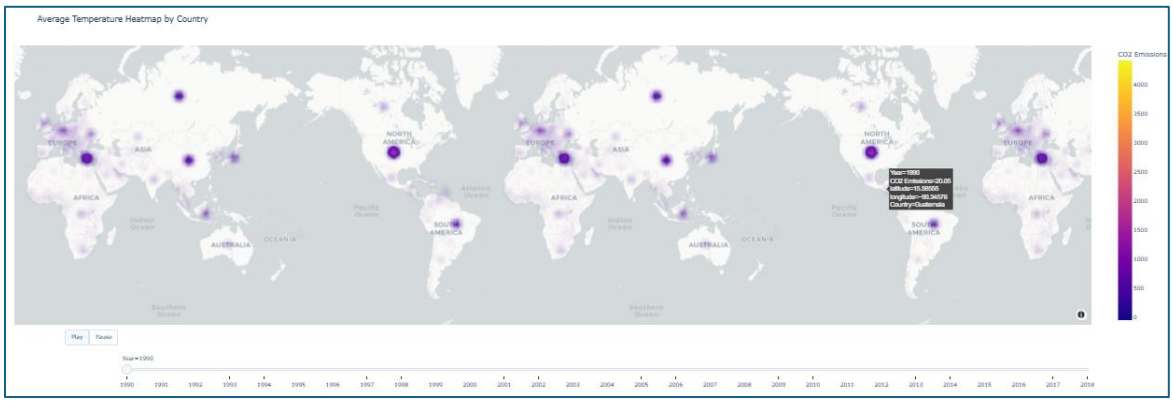
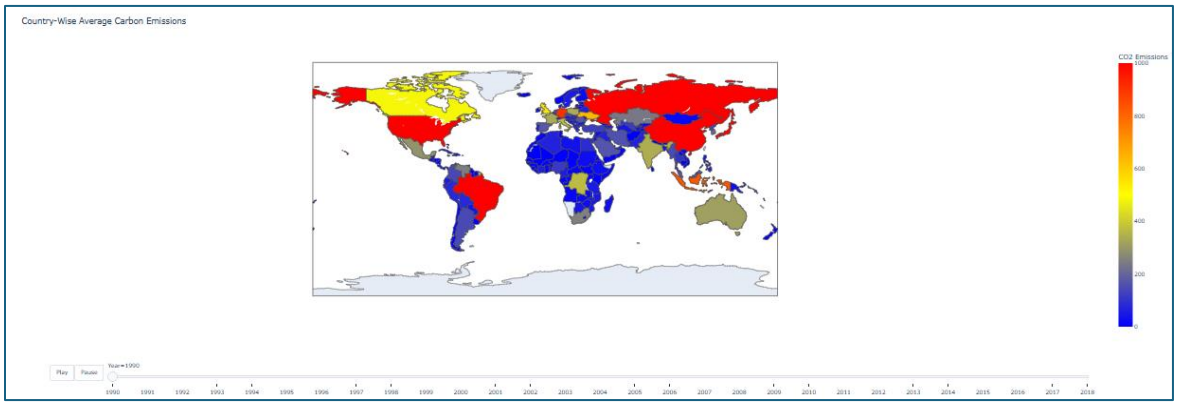


Fig 4: Choropleth Map

Carbon Emissions Visualizations:

- *Heat Maps & Bar Charts:* Highlighting regional and temporal variations in emissions, identifying top emitters, and tracing emission trajectories over decades.
- *Line Charts:* Tracking how both leading and lagging emitters evolve, clarifying whether global mitigation efforts are succeeding.





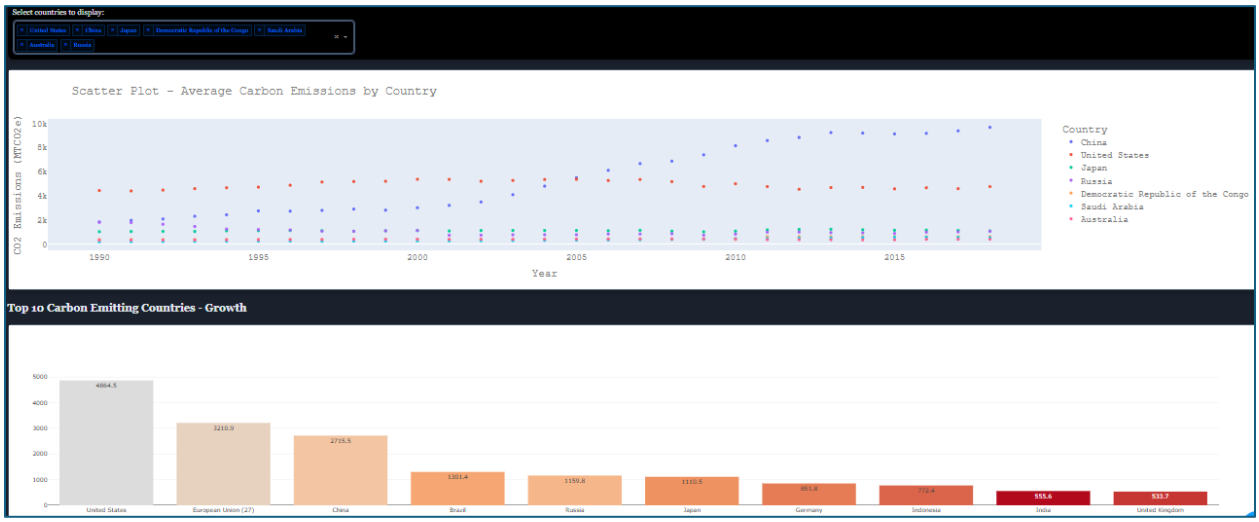
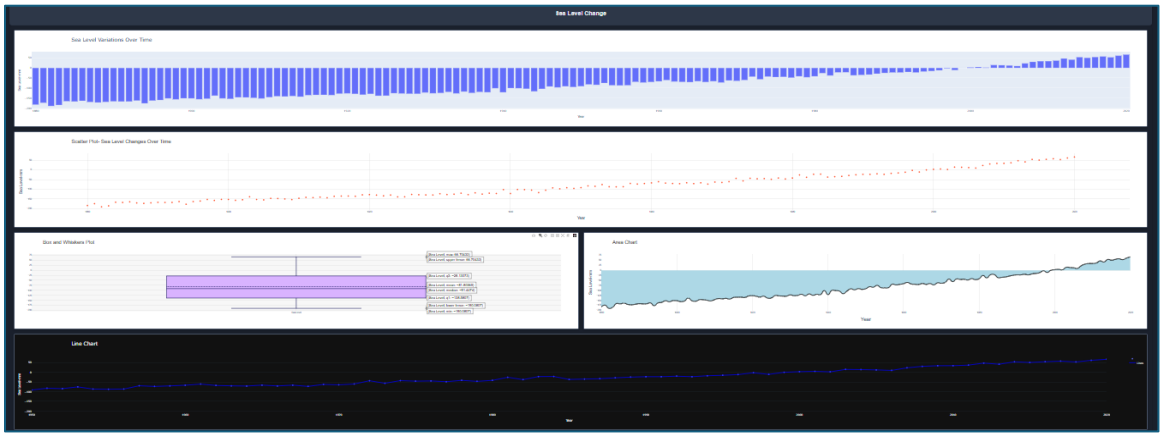


Fig 5: Illustrates Heatmaps, Scatter plot for the selected countries over the time.

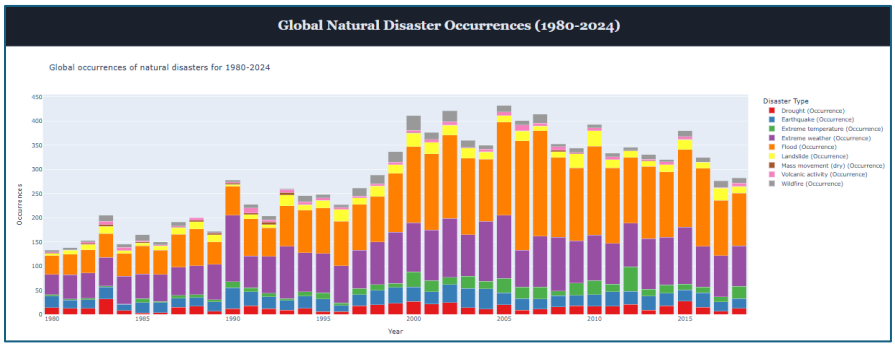
Sea Level Visualizations:

Bar, Scatter, and Box-and-Whisker Charts: Offering complementary views of sea-level changes from 1880 to 2020, helping to understand both the magnitude and variability of rise.



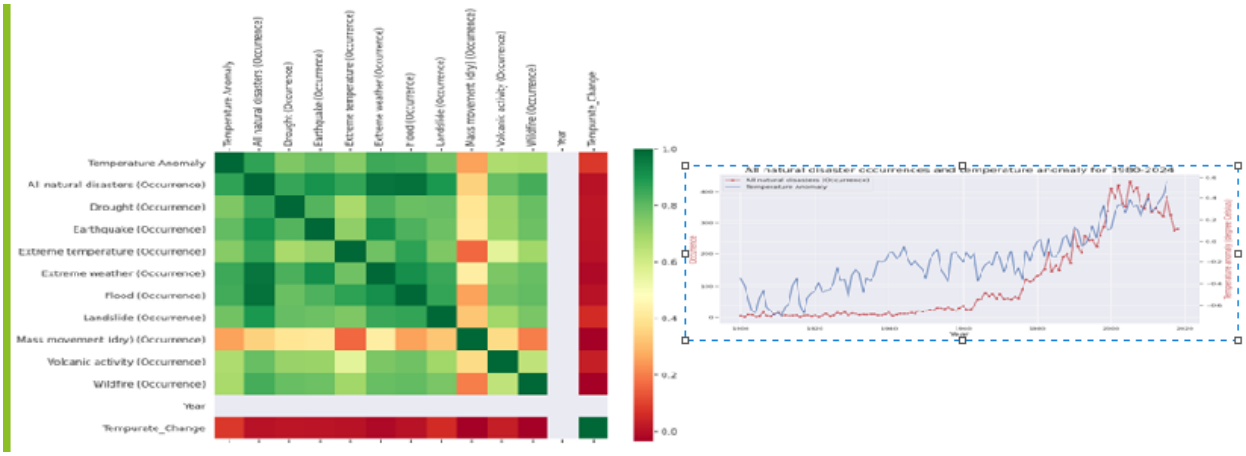
Natural Disaster Visualizations:

Stacked Bar Charts: Depicting the frequency of various natural disaster types over time, revealing trends in the prevalence of floods, droughts, and other events.



Correlations:

Line Plot and Heatmap**:** Illustrating the correlation between temperature anomalies, natural disasters, and economic damage from 1990 to 2018. The combination of plot types helps reveal both the trend in temperature anomalies and the strength of their relationship with disasters.



Results and Insights:
By integrating and visualizing multiple climate-related datasets, we derived key insights:

1. Correlations Between Carbon Emissions and Temperature**:** Historical emissions and temperature anomalies show a strong positive correlation, especially after the early 20th century, emphasizing the role of human activity in driving climate change.

2. Rising Sea Levels and Warming**:** Higher temperatures correlate with rising sea levels, with accelerated increases in both in recent decades, driven by ice melt and thermal expansion.

3. Increased Natural Disaster Frequency with Temperature Anomalies: A positive correlation between temperature anomalies and natural disasters, particularly floods and extreme weather, is evident since the mid-20th century.

4. Regional Disparities and Long-Term Trends**:** Choropleth maps reveal regions with disproportionately high emissions and temperature anomalies, guiding more targeted climate policies.

Challenges:

- Data Quality: Missing or incomplete data required interpolation, introducing uncertainty.
- Data Volume: Large datasets posed computational challenges, addressed through efficient preprocessing.
- Visualization Complexity: Designing intuitive multi-dimensional visualizations for non-linear relationships required iteration.

Future Work:

- Expanded Data Integration: Adding socioeconomic data (e.g., GDP, population density) for better understanding of climate impacts on human systems.
- Forecasting and Predictive Models**:** Incorporating machine learning for predictive climate modelling.

Conclusions:
Our platform integrates global climate data- temperature, emissions, sea-level rise, and natural disasters—into a unified environment, helping users uncover correlations and trends. This tool provides a comprehensive view, supporting evidence-based decision-making for climate policy. As data resources and predictive models evolve, the platform can serve as a powerful tool for guiding climate action and fostering resilient communities.

Source Code:
The file Climate_Analyzer.py is for running the interface.

References:
[1] IPCC AR6 Synthesis Report: Climate Change 2023.
[2] Liu, C., Chen, M., He, H. "Interactive Climate Data Analysis Dashboard," IEEE Transactions, 2019.
[3] Smith, P., Jones, R., Williams, J., Brown, K. "Linking Emissions and Sea-Level Rise," Climate Journal, 2021.
[4] NASA GISS, "Surface Temperature Analysis," 2024.
[5] Kaggle Datasets, "Berkeley Earth's Surface Temperature, Carbon Emissions, Sea Level Rise, and Natural Disasters," 2024.